4. Vehicle Rental System

Program :

```
from abc import ABC, abstractmethod


# Abstraction
class Vehicle(ABC):


    def __init__(self, days):
        self.days = days


    @abstractmethod
    def calculate_rent(self):
        pass


# Inheritance
class Car(Vehicle):


    def __init__(self, days):
        super().__init__(days)
        self.rate = 1000   # rent per day


    # Polymorphism
    def calculate_rent(self):
        return self.days * self.rate


class Bike(Vehicle):
```

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```
    def __init__(self, days):
        super().__init__(days)
        self.rate = 500   # rent per day

    # Polymorphism
    def calculate_rent(self):
        return self.days * self.rate

# ---- Main Program ----

vehicle_type = input("Enter Vehicle (Car/Bike): ")
days = int(input("Enter Number of Days: "))

if vehicle_type.lower() == "car":
    v = Car(days)
elif vehicle_type.lower() == "bike":
    v = Bike(days)
else:
    print("Invalid Vehicle Type")
    exit()

print("Total Rent:", v.calculate_rent())
```

Output :

Enter Vehicle (Car/Bike): Car

Enter Number of Days: 30

Total Rent: 30000

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